

BHASKAR SINGH

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<https://bhaskars-portfolio.vercel.app>

Education

Graphic Era Deemed to be University

Sep 2022 - July 2026

Bachelor of Technology (B.Tech) in Computer Science and Engineering (AI & ML)

CGPA: 8.24

Experience

Artificial Intelligence and IBM Cloud Technologies Intern

July 2025 – Aug 2025

IBM SkillsBuild Program

Remote

- Built and deployed an AI-powered chatbot on IBM Cloud to improve digital financial literacy for end users.
- Integrated a structured knowledge base that automated responses to user queries without human intervention.
- [GitHub Documentation](#)

Projects

Vision-Language Models for Electronic Meter Reading (Florence-2 & PaliGemma) — [GitHub](#)

Dec 2025

- Built a vision-language system to read numeric energy meter values from images.
- Fine-tuned Florence-2 as a baseline model, tracking training and validation loss for OCR-based meter reading.
- Used LoRA with PaliGemma (3B parameters) to fine-tune the model efficiently while keeping base weights frozen.
- Applied 4-bit quantization (BitsAndBytes) to reduce memory usage and verified results on unseen meter images

DSA GenAI Tutor (Retrieval-Augmented Generation) — [GitHub](#)

July 2025 - Nov 2025

- Designed and deployed a domain-specific GenAI tutor using a Retrieval-Augmented Generation (RAG) pipeline for DSA concepts.
- Built an automated ingestion pipeline converting 100+ open-source repositories into a vectorized semantic knowledge base using embeddings.

Sports Image Classification using CNNs — [GitHub](#) — [Live Demo](#)

Feb 2025

- Trained a transfer learning model (MobileNetV2) and a custom CNN on images spanning 100 sports categories.
- Achieved 94% training accuracy, 91% validation accuracy, and 89.9% test accuracy on unseen data.

Text Classification & Summarization (RNN + BART) — [GitHub](#) — [Live Demo](#)

Oct 2024 - Nov 2024

- Built a multi-class text classification system achieving 89.5% accuracy and Macro F1-score of 89%.
- Evaluated performance using ROC-AUC (0.9794) and Log Loss (0.3102) across four news categories.
- Fine-tuned a BART-based summarization model on the CNN/DailyMail dataset.
- Achieved ROUGE-1: 42.8, ROUGE-2: 19.6, and ROUGE-L: 39.6, demonstrating strong summarization quality.

Technical Skills

- Languages:** Java, Python
- Generative AI:** Large Language Models (LLMs), VLMs, RAG, Vector Databases, Fine-Tuning, LoRA
- Machine Learning:** ML Algorithms, Model Evaluation, Regularization, Hyperparameter Tuning
- Deep Learning:** CNN, RNN, LSTM, BiLSTM, Transformers (BERT, BART)
- Computer Vision:** VLMs, OCR, Image Understanding, Transfer Learning
- Natural Language Processing:** NLP Algorithms, Text Summarization, Question Answering, Prompt Engineering
- Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, HuggingFace, LangChain
- Databases:** MySQL

Certifications

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| AWS Certified Machine Learning Engineer - Associate | Credential | Dec 2025 |
| AWS Certified Cloud Practitioner | Credential | Sep 2024 |
| Microsoft Certified Azure AI Fundamentals | Credential | Dec 2023 |

Leadership & Achievements

- Solved 300+ problems on LeetCode and GeeksforGeeks, improving data structures & algorithms expertise.
- Organized and led multiple tech initiatives and workshops, driving AI/ML learning culture among 150+ students. Anchored and hosted various technical and cultural events.
- Collaborated with a 15-member core team, managing events, content, and project execution.